# Unit 4 :: Data Preparation and Management

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# Syllabus

## Unit 4: Data preparation and management

Basic pre-processing of data: filtering, sorting, normalization and handling missing values; managing multisheet workbooks and structured datasets, importing data from diverse file formats (CSV, ASCII, DAT), and ensuring interoperability between different spreadsheet and analysis platforms.

# 1 Importing data from diverse file formats

Suppose a physics experiment records a variable which is of interest to us, such as temperature, pressure, voltage, count rate etc. These variables are recorded (or sampled) at a sequence of time instants. Now this raw record of data produced by an instrument is almost certainly unfit for direct analysis because

a) It may contain unreliable readings.

b) It may be out of chronological order.

c) It may mix standard and non-standard units.

d) It may contain gaps.

Then we say that such a dataset must be *pre-processed* before it can be visualised or fitted to a theoretical curve.

For the purpose of pre-processing, the dataset must be first stored and archived in the form of a file. Three file formats are widely physics laboratory exports:

1. Comma-separated values (CSV files)
2. Instrument-specific data (DAT files)
3. Microsoft Excel spreadsheets (XLSX files)

## 1.1 ASCII files

ASCII, which is an acronym for *American Standard Code for Information Interchange*, is a character encoding standard. Originally, it was used for representing a particular set of 95 printable characters and 33 control characters, thereby giving a total of 128 code points, using 7 bits. Later, an 8-bit encoding extended the ASCII set to a total of 256 characters.

An ASCII file is one that contains data made up of ASCII characters, i.e. it is essentially raw text. Each byte in the file contains one character that conforms to the standard ASCII code.

Source codes of computer programs, batch files, macros and scripts are straight text and stored as ASCII files. HTML and XML files are also ASCII files. Text editors such as Notepad create ASCII files as their native file format.

CSV and DAT files are two of the most commonly used ASCII file formats used for storing scientific data for subsequent analysis.

### 1.1.1 CSV files

A comma-separated values (CSV) file is a plain text data format for storing tabular data where the fields (values) of a record are separated by a comma and each record is a line (i.e. newline separated). For example, the following are the contents of a CSV file:

```
id, name, email
1, John, john.doe@example.com
2, Jane, janey72@test.org
```

In software that generally deals with tabular data such as a database or a spreadsheet, the contents of the above CSV file is parsed in the form of a table as follows:

| **id** | **name** | **email** |
|---|---|---|
| 1 | John | john.doe@example.com |
| 2 | Jane | janey72@test.org |

The first line is used as the table header and the remaining lines are used as the rows of the table.

In Origin

#### 1.1.2 DAT files

# 2 Insert title

Insert content.

# 3 Insert title

Insert content.

# 4 Insert title

Insert content.

# 5 Insert title

Insert content.

# 6 Insert title

Insert content.